

TRACK B

SHADOW AI GOVERNANCE

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EXECUTIVE SUMMARY

Modern enterprise security architectures are critically exposed by Shadow AI; the unauthorized use of consumer LLMs by employees handling sensitive corporate data. While legacy compliance suites (like Microsoft Purview and Defender) rely on heavy platform ecosystem lock-ins and frustrating binary blocks, VerifAI disrupts this space with a lightweight, cross-platform enterprise browser extension.

By operating entirely within the browser runtime across all major Chromium, Gecko, and WebKit engines, VerifAI eliminates prohibitive licensing costs. Rather than triggering user friction with standard access blocks that drive employees to personal devices, VerifAI deploys Just-In-Time (Jit-Route) Prompt Interception. When a user attempts to submit data to an unapproved AI frontend, the extension instantly scans the DOM and smoothly redirects the productivity intent directly to corporate-sanctioned, secure LLM alternatives.

Can be installed as a standalone browser plug-in or managed securely via Enterprise MDM deployment paths (such as JAMF and Microsoft Intune) to prevent tampering and local uninstalls, VerifAI channels data into a dual-engine Security Operations Center (SOC) dashboard. This grants security teams visibility across browser process executions and browser-level proxy redirections, reclaiming data governance without stifling workforce productivity.

VERIFAI PRODUCT BLUEPRINT COMPARISON TABLE

EVALUATION ATTRIBUTE	LEGACY ENTERPRISE SUITES (E.G., MICROSOFT E5)	VERIFAI ENTERPRISE SOLUTION
Architectural Scope	Windows, Edge, and Azure native lock-in.	Browser-native, cross-platform (Chrome, Edge, Brave, Firefox).
OS Footprint	Fragile or non-existent on MacOs, Linux, iOS, and Android.	OS-agnostic enforcement via unified browser runtime extension.
Enforcement Protocol	Binary hard blocks causing high user friction and bypass motivation.	Just-In-Time (Jit-Route) prompt interception and soft redirection.
Data Visibility	Monolithic logs requiring complex, high-tier compliance licensing.	Dual-engine SOC view tracking AI Activities and PII Redactions
Tamper Resistance	Dependent on heavy host-level kernel drivers.	Controlled via standardized MDM policies (force installed rules).

PROBLEM STATEMENT

Legacy Enterprise Suites (such as Microsoft E5) enforce a rigid, platform-locked architecture that creates severe operational vulnerabilities, high user friction, and inflated compliance costs across modern, heterogeneous IT environments.

CORE CHALLENGES

- **Platform Incompatibility:** Security coverage remains fragile or entirely non-existent on macOS, Linux, iOS, and Android due to a native dependency on Windows, Edge, and Azure ecosystems.
- **Operational Friction:** Enforcement protocols rely on disruptive, binary hard blocks that impede productivity and motivate users to actively bypass security controls.
- **Visibility Barriers:** Accessing critical data requires navigating monolithic logs buried behind complex, cost-prohibitive licensing tiers.
- **System Instability:** Tamper resistance is heavily dependent on intrusive, host-level kernel drivers that complicate endpoint stability and management.

CURRENT STATE

USER MOTIVATION	THE DILEMMA	RESULTING ACTION	CORPORATE RISK CREATED
Productivity Pressure	Fast delivery vs. Waiting for IT vetting	Employee uses free web-based LLMs	Invisible data traffic escaping network logs
Tool Gap / Lack of Features	Inefficient legacy software vs. High-speed AI capabilities	Employee inputs corporate data to generate code/summaries	Data leakage as proprietary info trains public models
Policy Ambiguity	Lack of official rules vs. High availability of consumer tools	Employee self-regulates and adopts external plugins	Compliance violations regarding privacy and data retention

SOLUTION

Help employees choose approved AI tools, reduce sensitive-data exposure, and give the SOC visibility without collecting raw prompts that can be installed in any browser.

WORKFLOW SUMMARY

Standalone Browser Plug-In

Zone	Key Components	Role in Workflow
User & Browser	Navigation, Network Rules, Blocks, Approved Agents	Initiates visits, evaluates compliance rules, and hosts allowed AI assistants.
Extension Logic	Block Page, MV3 Background, Events Queue	Intercepts policy violations, handles host-only events, and buffers telemetry safely.
Local Services	127.0.0.1 Service, Event File, Reporting	Authenticates delivery, writes persistent logs, and powers the offline dashboard.

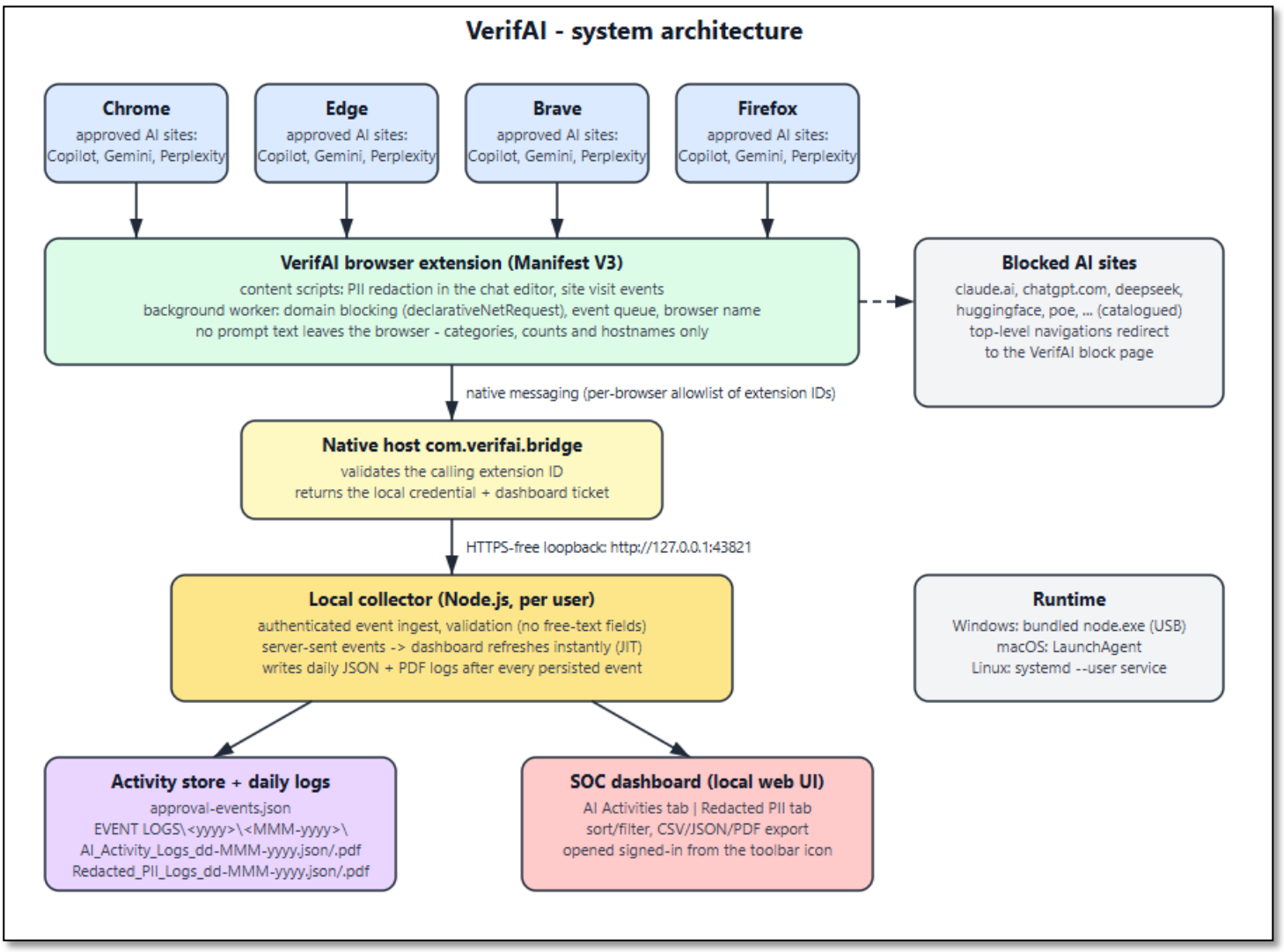
HOW IT WORKS

TECHNOLOGY USED

- **Platform used for building:** Github Copilot, VS Code Pro
- **Agents used:**

  Claude Fable 5.1

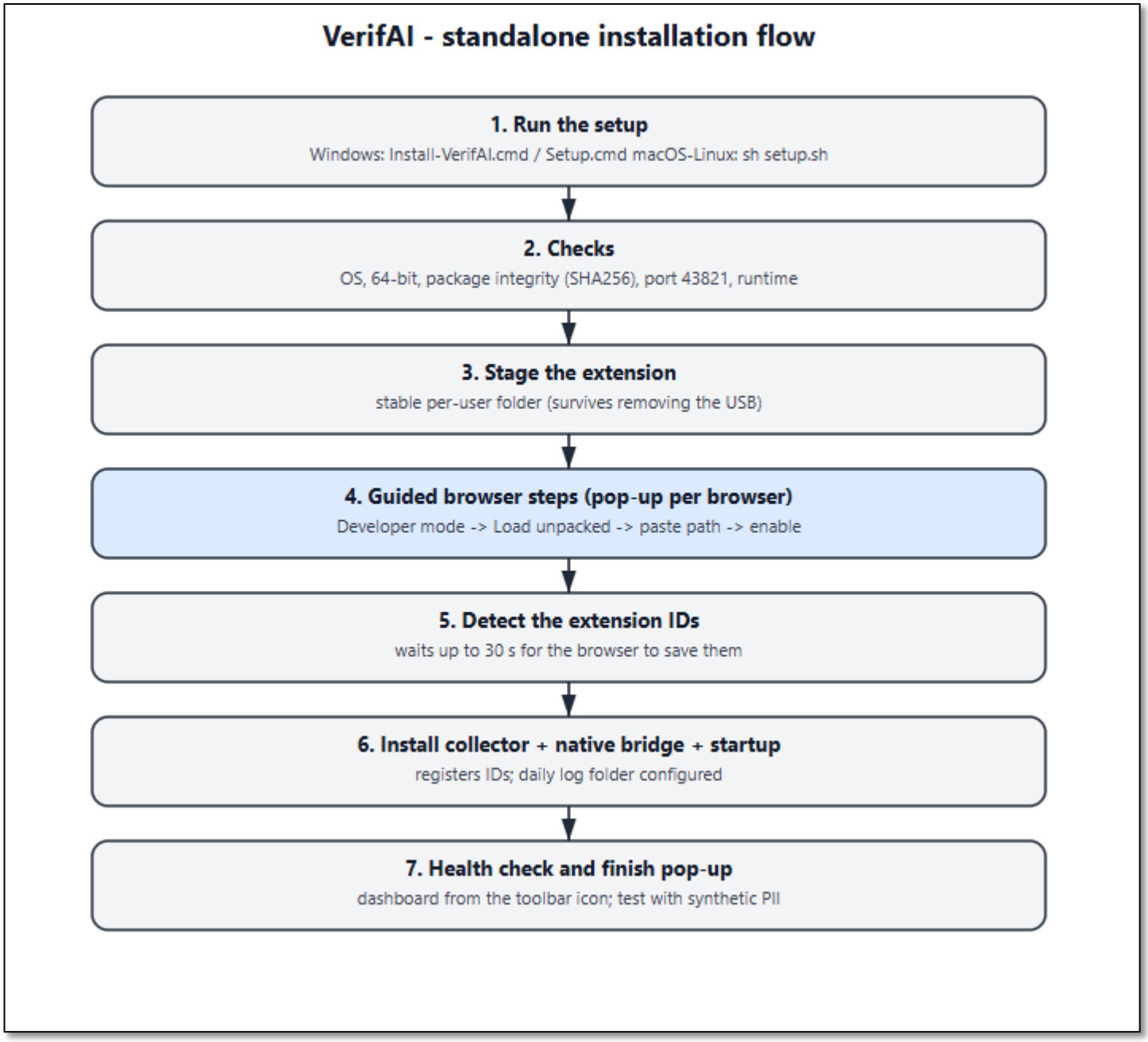
  GPT-6 Luna
- **Backend:** Typescript, Javascript, Vite
- **Frontend**: HTML5, CSS4
- **Deployment**: Download from the shared Google Drive



OPERATIONAL CONTEXT

COMPONENT	EMBEDDED SUB-TEXT DETAIL	ARCHITECTURAL PURPOSE
User Site Navigation	Triggers policy evaluation	The starting trigger where the user types in a shadow AI in a browser.
Browser Network Rules	Declarative Net Request	Utilizes high-performance Manifest V3 APIs to evaluate safety policies natively.
Extension Block Page	UI blocking overlay	Informs the employee of a violation while presenting safe, approved alternatives.
MV3 Background	Service worker mesh	The persistent orchestrator running out-of-process to manage queues and syncs.
Events & Pending Queue	IndexedDB backup storage	Implements local browser persistence to prevent data loss during network drops.
Local Service	Native messaging host	Runs locally on the loopback address (127.0.0.1) to pipe logs directly to storage.
Local Event File	Encrypted JSONL logs	Secures captured telemetry directly on the physical storage layer.
Dashboard UI	React/HTML5 frontend	Delivers a clean, zero-external-dependency viewer for SOC/risk management teams.

VERIFAI STANDALONE INSTALLATION FLOW



VERIFAI JIT REDACTION AND ACTIVITY FLOW

